HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Supported the **Permanent Way design team** by analysing track geometry data to inform engineering calculations and the production of **project drawings**.
- Contributed to the drafting of technical specifications and project reports, ensuring clarity and adherence to engineering best practices for **rail infrastructure projects**.
- Assisted with **cross-disciplinary coordination**, integrating design elements from structural and signalling teams to enable fully cohesive project designs using **latest software tools**.
- Performed **data analysis** on sensor readings from existing rail lines to identify patterns and support predictive maintenance strategies, impacting a team of **8 engineers**.
- Developed **Python scripts** to automate the processing of large datasets related to track condition and performance, reducing manual data handling time by **40%**.
- Containerised **Java Spring Boot** microservices for a financial operations dashboard using **Docker**, deploying them to an **AWS EC2** environment.
- Collaborated with operations teams on a **payments operations dashboard**, performing root cause analysis for P1 incidents and managing SLAs within a **production environment**.

SKILLS

Rail Engineering – Permanent Way Design, Rail Infrastructure, Track Geometry, Sustainable Transport, Railway Engineering

Software Development – Java, Spring Boot, Python, REST APIs, Docker, Git, Jenkins

Data Analysis & BI – SQL, PostgreSQL, MySQL, Oracle, pandas, NumPy, Power BI, Tableau, DAX, Power Query

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Linux, Shell Scripting, CI/CD

Enterprise Systems – ERP Integration, Change Management, Jira, Confluence, Dynatrace, Splunk

PROJECTS

Rail Infrastructure Data Analysis Platform

- Developed a **Python-based data processing pipeline** to ingest and clean track geometry and sensor data for a large-scale rail expansion project.
- Engineered **SQL queries** across PostgreSQL and Oracle databases to extract and aggregate performance metrics for design calculations, supporting a team of **12 engineers**.
- Built **interactive dashboards in Power BI** to visualise track condition trends and highlight areas requiring immediate engineering attention.

Automated Reporting for Enterprise Systems

- Created **Python scripts** to automate the generation of daily operational reports from multiple enterprise systems, reducing manual effort by approximately **30%**.
- Integrated data extraction logic with **Java Spring Boot** services to provide real-time data feeds for internal business intelligence tools.
- Wrote **SQL scripts** for data validation and transformation prior to report generation, ensuring data integrity across interconnected systems.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)